

```
!pip install -q sentence-transformers scikit-learn pandas torch
```

```
import pandas as pd
from sentence_transformers import CrossEncoder
```

```
from datasets import load_dataset

dataset = load_dataset("microsoft/ms_marco", "v1.1")
```

```
README.md:          9.48k/? [00:00<00:00, 850kB/s]

v1.1/validation-00000-of-00001.parquet: 100%          21.4M/21.4M [00:03<00:00, 11.7MB/s]

v1.1/train-00000-of-00001.parquet: 100%              175M/175M [00:01<00:00, 84.2MB/s]

v1.1/test-00000-of-00001.parquet: 100%              20.5M/20.5M [00:01<00:00, 18.8MB/s]

Generating validation split: 100%                    10047/10047 [00:00<00:00, 33758.71 examples/s]

Generating train split: 100%                         82326/82326 [00:01<00:00, 63394.86 examples/s]

Generating test split: 100%                          9650/9650 [00:00<00:00, 38682.21 examples/s]
```

```
print(dataset)
```

```
DatasetDict({
  validation: Dataset({
    features: ['answers', 'passages', 'query', 'query_id', 'query_type', 'wellFormedAnswers'],
    num_rows: 10047
  })
  train: Dataset({
    features: ['answers', 'passages', 'query', 'query_id', 'query_type', 'wellFormedAnswers'],
    num_rows: 82326
  })
  test: Dataset({
    features: ['answers', 'passages', 'query', 'query_id', 'query_type', 'wellFormedAnswers'],
    num_rows: 9650
  })
})
```

```
print(dataset["train"][0])
```

```
{'answers': ['Results-Based Accountability is a disciplined way of thinking and taking action that communities can use to im
```

```
dataset["train"][0]["passages"]
```

```
{'is_selected': [0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0],
 'passage_text': ["Since 2007, the RBA's outstanding reputation has been affected by the 'Securency' or NPA scandal.
These RBA subsidiaries were involved in bribing overseas officials so that Australia might win lucrative note-printing
contracts. The assets of the bank include the gold and foreign exchange reserves of Australia, which is estimated to have
a net worth of A$101 billion. Nearly 94% of the RBA's employees work at its headquarters in Sydney, New South Wales and
at the Business Resumption Site.",
"The Reserve Bank of Australia (RBA) came into being on 14 January 1960 as Australia 's central bank and banknote
issuing authority, when the Reserve Bank Act 1959 removed the central banking functions from the Commonwealth Bank. The
assets of the bank include the gold and foreign exchange reserves of Australia, which is estimated to have a net worth of
A$101 billion. Nearly 94% of the RBA's employees work at its headquarters in Sydney, New South Wales and at the Business
Resumption Site.",
'RBA Recognized with the 2014 Microsoft US Regional Partner of the ... by PR Newswire. Contract Awarded for supply and
support the. Securitisations System used for risk management and analysis. ',
'The inner workings of a rebuildable atomizer are surprisingly simple. The coil inside the RBA is made of some type of
resistance wire, normally Kanthal or nichrome. When a current is applied to the coil (resistance wire), it heats up and
the heated coil then vaporizes the eliquid. 1 The bottom feed RBA is, perhaps, the easiest of all RBA types to build,
maintain, and use. 2 It is filled from below, much like bottom coil clearomizer. 3 Bottom feed RBAs can utilize cotton
instead of silica for the wick. 4 The Genesis, or genny, is a top feed RBA that utilizes a short woven mesh wire.',
'Results-Based Accountability® (also known as RBA) is a disciplined way of thinking and taking action that communities
can use to improve the lives of children, youth, families, adults and the community as a whole. RBA is also used by
organizations to improve the performance of their programs. RBA improves the lives of children, families, and communities
and the performance of programs because RBA: 1 Gets from talk to action quickly; 2 Is a simple, common sense process
that everyone can understand; 3 Helps groups to surface and challenge assumptions that can be barriers to innovation;',
'Results-Based Accountability® (also known as RBA) is a disciplined way of thinking and taking action that communities
can use to improve the lives of children, youth, families, adults and the community as a whole. RBA is also used by
organizations to improve the performance of their programs. Creating Community Impact with RBA. Community impact focuses
on conditions of well-being for children, families and the community as a whole that a group of leaders is working
collectively to improve. For example: "Residents with good jobs," "Children ready for school," or "A safe and clean
neighborhood".',
'RBA uses a data-driven, decision-making process to help communities and organizations get beyond talking about
```

problems to taking action to solve problems. It is a simple, common sense framework that everyone can understand. RBA starts with ends and works backward, towards means. The “end” or difference you are trying to make looks slightly different if you are working on a broad community level or are focusing on your specific program or organization. RBA improves the lives of children, families, and communities and the performance of programs because RBA: 1 Gets from talk to action quickly; 2 Is a simple, common sense process that everyone can understand; 3 Helps groups to surface and challenge assumptions that can be barriers to innovation;’,

’vs. NetIQ Identity Manager. Risk-based authentication (RBA) is a method of applying varying levels of stringency to authentication processes based on the likelihood that access to a given system could result in its being compromised. Risk-based authentication can be categorized as either user-dependent or transaction-dependent. User-dependent RBA processes employ the same authentication for every session initiated by a given user; the exact credentials that the site demands depend on who the user is.’,

’A rebuildable atomizer (RBA), often referred to as simply a “rebuildable,” is just a special type of atomizer used in the Vape Pen and Mod Industry that connects to a personal vaporizer. 1 The bottom feed RBA is, perhaps, the easiest of all RBA types to build, maintain, and use. 2 It is filled from below, much like bottom coil clearomizer. 3 Bottom feed RBAs can utilize cotton instead of silica for the wick. 4 The Genesis, or genny, is a top feed RBA that utilizes a short woven mesh wire.’,

’Get To Know Us. RBA is a digital and technology consultancy with roots in strategy, design and technology. Our team of specialists help progressive companies deliver modern digital experiences backed by proven technology engineering. ’],

’url’: [https://en.wikipedia.org/wiki/Reserve_Bank_of_Australia’,
https://en.wikipedia.org/wiki/Reserve_Bank_of_Australia’,
<http://acronyms.thefreedictionary.com/RBA>’,
<https://www.slimvapepen.com/rebuildable-atomizer-rba/>’,
<http://rba-africa.com/about/what-is-rba/>’,
<http://resultsleadership.org/what-is-results-based-accountability-rba/>’,
<http://rba-africa.com/about/what-is-rba/>’,
<http://searchsecurity.techtarget.com/definition/risk-based-authentication-RBA>’,
<https://www.slimvapepen.com/rebuildable-atomizer-rba/>’,
<http://www.rbaconsulting.com/>’]

```
rows = []

for item in dataset["train"]:
    query = item["query"]
    answers = item["answers"]

    for p in item["passages"]["passage_text"]:
        rows.append({
            "query": query,
            "finalpassage": p,
            "answers": answers
        })
```

```
import pandas as pd
```

```
df = pd.DataFrame(rows)
print(df.head())
```

	query	finalpassage \	answers
0	what is rba	Since 2007, the RBA's outstanding reputation h...	[Results-Based Accountability is a disciplined...
1	what is rba	The Reserve Bank of Australia (RBA) came into ...	[Results-Based Accountability is a disciplined...
2	what is rba	RBA Recognized with the 2014 Microsoft US Regi...	[Results-Based Accountability is a disciplined...
3	what is rba	The inner workings of a rebuildable atomizer a...	[Results-Based Accountability is a disciplined...
4	what is rba	Results-Based Accountability® (also known as R...	[Results-Based Accountability is a disciplined...

```
from sklearn.model_selection import train_test_split

train_df, valid_df = train_test_split(
    df,
    test_size=0.2,
    random_state=42
)
```

```
train_df.to_csv("train.csv", index=False)
valid_df.to_csv("valid.csv", index=False)
```

```
print(pd.read_csv("train.csv").head())
print(pd.read_csv("valid.csv").head())
```

	query \
0	meaning of the name noma

```

1 how did the civil rights act impact the civil ...
2                 cape ann distance from boston
3                 what is skin rash
4                 accounting process definition

                                finalpassage \
0 : a spreading invasive gangrene chiefly of the...
1 CIVIL RIGHTS MOVEMENT. The civil rights moveme...
2 The total driving distance from Boston, MA to ...
3 A rash may be localized in one part of the bod...
4 Definition. A sequence of activities involving...

                                answers
0                                ['Fate']
1 ['The Civil Rights Act of 1964, which ended se...
2                                ['30 Miles']
3 ['It is a noticeable change in the texture or ...
4 ['A series of activities that begins with a tr...
                                query \
0 how to use headphones with hdmi tv
1 the muscle term fascia means quizlet
2 the year penicillin was discovered
3 pay harvard tuition
4 what is prpp

                                finalpassage \
0 In this video I show you how to use head phone...
1 subcutaneous fascia. a continuous layer of con...
2 The penicillin was discovered by Alexander Fle...
3 As you can see, additional costs aside from tu...
4 The Pooled Registered Pension Plan (PRPP). A P...

                                answers
0 ['Make sure the appropriate devices are connec...
1 ['A sheet or band of fibrous connective tissue...
2                                ['1928', '1928', '1928']
3 ['The total 2015-2016 cost of attending Harvar...
4 ['A PRPP is a new, accessible, straightforward...
```

```

rows = []

for item in dataset["train"]:
    query = item["query"]
    answers = item["answers"]

    passages = item["passages"]["passage_text"]
    labels = item["passages"]["is_selected"]

    for p, l in zip(passages, labels):
        rows.append({
            "query": query,
            "finalpassage": p,
            "answers": answers,
            "label": int(l)
        })
```

```

df = pd.DataFrame(rows)
print(df["label"].value_counts())
```

```

label
0    587670
1     88523
Name: count, dtype: int64
```

```

from sklearn.model_selection import train_test_split

train_df, valid_df = train_test_split(
    df,
    test_size=0.2,
    random_state=42,
    stratify=df["label"]
)
```

```

def clean_text_columns(df, query_col, text_col):
    df = df.dropna(subset=[query_col, text_col])
    df[query_col] = df[query_col].astype(str)
    df[text_col] = df[text_col].astype(str)
    return df
```

```

QUERY_COL = "query"
TEXT_COL = "finalpassage"
ANSWER_COL = "answers"
LABEL_COL = "label" # only if using labels

```

```
valid_df = clean_text_columns(valid_df, QUERY_COL, TEXT_COL)
```

```

# Column names
QUERY_COL = "query"
TEXT_COL = "finalpassage"
ANSWER_COL = "answers"
LABEL_COL = "label"

# Cleaning function
def clean_text_columns(df, query_col, text_col):
    df = df.dropna(subset=[query_col, text_col])
    df[query_col] = df[query_col].astype(str)
    df[text_col] = df[text_col].astype(str)
    return df

```

```

print(valid_df.head())
print(valid_df.isnull().sum())

```

```

              query \
317720              laying a path cost
558233 why do geologists find iceland a useful place ...
494196              where is steyn city
314018              avg temp for carmel
555576 do piers morgan and howie mandel really dislik...

              finalpassage \
317720 1 This Old House explains how to lay a gravel ...
558233 Scientists might find Iceland a good place to ...
494196 WATCH: Official opening of Steyn City. INSURAN...
314018 The highest average temperature in Carmel is J...
555576 So I think it's just a little competition betw...

              answers  label
317720 [$100 -$1,000 for gravel pathway.$6 and $12 pe...    0
558233 [Because of large amounts of lava extrusion, t...    0
494196 [On the edge of Sandton, connecting Fourways t...    0
314018 [50.4 degrees.]    0
555576 [yes]    1
query    0
finalpassage 0
answers    0
label    0
dtype: int64

```

```

from sentence_transformers import CrossEncoder

reranker = CrossEncoder("cross-encoder/ms-marco-MiniLM-L-6-v2")

```

```

Loading weights: 100% 105/105 [00:00<00:00, 497.99it/s, Materializing param=classifier.weight]
BertForSequenceClassification LOAD REPORT from: cross-encoder/ms-marco-MiniLM-L-6-v2
Key          | Status      | |
-----+-----+---+
bert.embeddings.position_ids | UNEXPECTED | |

Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.

```

```

def rerank(df):
    pairs = list(zip(df[QUERY_COL], df[TEXT_COL]))
    scores = reranker.predict(pairs)

    df = df.copy()
    df["rerank_score"] = scores

    return df.sort_values("rerank_score", ascending=False)

```

```
from sentence_transformers import CrossEncoder
```

```
reranker = CrossEncoder(
    "cross-encoder/ms-marco-MiniLM-L-6-v2",
    device="cuda"
)
```

Loading weights: 100% 105/105 [00:00<00:00, 461.22it/s, Materializing param=classifier.weight]

BertForSequenceClassification LOAD REPORT from: cross-encoder/ms-marco-MiniLM-L-6-v2

Key	Status	
bert.embeddings.position_ids	UNEXPECTED	

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.

```
import pandas as pd
from sentence_transformers import CrossEncoder
```

```
QUERY_COL = "query"
TEXT_COL = "finalpassage"
ANSWER_COL = "answers"
LABEL_COL = "label"
```

```
reranker = CrossEncoder(
    "cross-encoder/ms-marco-MiniLM-L-6-v2",
    device="cuda"
)
```

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t

Loading weights: 100% 105/105 [00:00<00:00, 386.21it/s, Materializing param=classifier.weight]

BertForSequenceClassification LOAD REPORT from: cross-encoder/ms-marco-MiniLM-L-6-v2

Key	Status	
bert.embeddings.position_ids	UNEXPECTED	

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.

```
def rerank(df):
    pairs = list(zip(df[QUERY_COL], df[TEXT_COL]))
    scores = reranker.predict(pairs, batch_size=32)

    df = df.copy()
    df["rerank_score"] = scores
    return df.sort_values("rerank_score", ascending=False)
```

```
reranked_valid = rerank(valid_df)
```

```
print("Reranking completed!")
print(reranked_valid.shape)
```

```
Reranking completed!
(135239, 5)
```

```
assert "reranked_valid" in globals(), "Please run reranked_valid = rerank(valid_df) first"
print("reranked_valid shape:", reranked_valid.shape)
```

```
reranked_valid shape: (135239, 5)
```

```
reranked_valid[[QUERY_COL, TEXT_COL, "rerank_score"]].head(10)
```

	query	finalpassage	rerank_score
322502	what is a vertebral artery dissection	Vertebral artery dissection (abbreviated VAD, ...	11.640962
131588	professional golfer average salary	Average Professional Golfer Salaries. The aver...	11.603867
479763	what is quality assurance	Quality assurance (QA) is a process-centered a...	11.602543
477369	what is soy sauce	Soy sauce (also called soya sauce) is a condim...	11.602080
199498	what is KNOWLEDGE management	Knowledge management is the name of a concept ...	11.601532
492461	what is enabling technology	From Wikipedia, the free encyclopedia. An enab...	11.580256
654249	what is c-reactive protein	C-reactive protein (CRP) is an annular (ring-s...	11.577459
653856	what is an indemnity agreement	What Is An Indemnity Agreement? □ An indemnity...	11.573626
457724	what is a vinaigrette	A vinaigrette (/ˈvɪnəˈɡrɛt/ vin-ə-GRET) is a cu...	11.573234
46242	average salary for rural carrier associate	Average Rural Carrier Associate Salaries. The ...	11.569812

```
reranked_by_query = (
    reranked_valid
    .groupby(QUERY_COL, group_keys=False)
    .apply(lambda x: x.sort_values("rerank_score", ascending=False))
)
```

/tmp/ipython-input-3003154688.py:4: DeprecationWarning: DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated. Use DataFrameGroupBy.apply_grouped instead.

```
.apply(lambda x: x.sort_values("rerank_score", ascending=False))
```

```
TOP_K = 5
topk_results = reranked_by_query.groupby(QUERY_COL).head(TOP_K)
```

```
sample_query = topk_results[QUERY_COL].iloc[0]

topk_results[topk_results[QUERY_COL] == sample_query][
    [TEXT_COL, "rerank_score"]
]
```

	finalpassage	rerank_score
322502	Vertebral artery dissection (abbreviated VAD, ...	11.640962
322500	Vertebral artery dissection. Dr Henry Knipe ● ...	10.236000
322496	Vertebral artery dissection (VAD) is a relativ...	9.759173

```
HAS_LABELS = LABEL_COL in topk_results.columns
print("Labels available:", HAS_LABELS)
```

Labels available: True

```
if HAS_LABELS:
    precision_k = (
        topk_results
        .groupby(QUERY_COL)[LABEL_COL]
        .mean()
        .mean()
    )
    print(f"Precision@{TOP_K}:", precision_k)
```

Precision@5: 0.13423756167963646

```
def reciprocal_rank(df):
    for idx, row in enumerate(df.itertuples(), start=1):
        if getattr(row, LABEL_COL, 0) == 1:
            return 1 / idx
    return 0
```

```
if HAS_LABELS:
    mrr = (
        reranked_by_query
```

```

        .groupby(QUERY_COL)
        .apply(reciprocal_rank)
        .mean()
    )
    print("MRR:", mrr)

```

MRR: 0.2246649238947817

/tmp/ipython-input-2567263654.py:5: DeprecationWarning: DataFrameGroupBy.apply operated on the grouping columns. This behavior will be removed in a future version. You can avoid this warning by using DataFrameGroupBy.apply_grouped.
.apply(reciprocal_rank)

```

import numpy as np

def ndcg_at_k(df, k):
    rels = df[LABEL_COL].tolist()
    dcg = sum(
        rel / np.log2(i + 2)
        for i, rel in enumerate(rels[:k])
    )
    ideal = sum(
        rel / np.log2(i + 2)
        for i, rel in enumerate(sorted(rels, reverse=True)[:k])
    )
    return dcg / ideal if ideal > 0 else 0

```

```

if HAS_LABELS:
    ndcg = (
        reranked_by_query
        .groupby(QUERY_COL)
        .apply(lambda x: ndcg_at_k(x, TOP_K))
        .mean()
    )
    print(f"NDCG@{TOP_K}:", ndcg)

```

NDCG@5: 0.23192898229961065

/tmp/ipython-input-3077513138.py:5: DeprecationWarning: DataFrameGroupBy.apply operated on the grouping columns. This behavior will be removed in a future version. You can avoid this warning by using DataFrameGroupBy.apply_grouped.
.apply(lambda x: ndcg_at_k(x, TOP_K))

```

topk_results.to_csv("final_reranked_results.csv", index=False)
print("Saved: final_reranked_results.csv")

```

Saved: final_reranked_results.csv

```

print("==== FINAL SUMMARY ===")
print("Total pairs:", len(reranked_valid))
print("Queries:", reranked_valid[QUERY_COL].nunique())
print("Average rerank score:", reranked_valid["rerank_score"].mean())

if HAS_LABELS:
    print(f"Precision@{TOP_K}:", precision_k)
    print("MRR:", mrr)
    print(f"NDCG@{TOP_K}:", ndcg)

```

```

==== FINAL SUMMARY ====
Total pairs: 135239
Queries: 68364
Average rerank score: 4.592267
Precision@5: 0.13423756167963646
MRR: 0.2246649238947817
NDCG@5: 0.23192898229961065

```